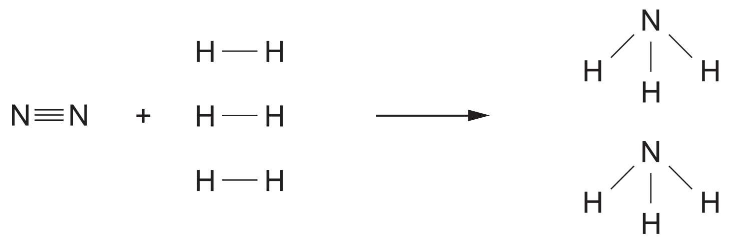
7. (a) The formation of ammonia can be represented by the following equation.



- The total amount of energy released in making the bonds in the products is 2340 kJ
- The total amount of energy released in making the bonds in the products is 94 kJ **more than** the total energy used in breaking the bonds in the reactants
- The amount of energy used to break the N≡N bond is 941 kJ

Use this information to calculate the energy used to break **one** H—H bond. [3]

Energy = ... kJ

Examiner only

(b) The Haber process used to make ammonia is usually carried out at a temperature of around 400 °C. Explain why this is the optimum temperature. [2]

A higher temperature is not used because ....

....

A lower temperature is not used because ....

....

(c) Ammonia is used in the production of fertilisers such as ammonium nitrate.

Give the balanced symbol equation for the reaction between nitric acid and ammonia to form ammonium nitrate. [2]

....

(d) Eutrophication is a problem caused by fertilisers being washed into waterways leading to overgrowth of plants. Describe how eutrophication leads to the death of aquatic organisms. [2]

....

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